Reg. No. | | | | | | | | | | | [13CE55]

FIFTH SEMESTER B.TECH/B.E EXAMINATION, DECEMBER-2015

HYDROLOGY

Time: 3 Hours **Maximum Marks:70**

Instruction: Answer any **FIVE** full Questions.

1. a. List and briefly explain various practical applications of Hydrology. **07**

b. A lake has an area of 10 Km^2. Observation of hydrological variables during 2014 has shown that : **07**

i. Precipitation=700 mm/year

ii. Average Inflow to lake=1.4m^3/sec

iii. Average outflow from lake=1.6m^3/sec Assume that there is no net water exchange between the lake and the ground water determine evaporation during 2014 using water-budget equation.

2. a. Explain different types of precipitation and distinguish between cold front and warm front. **08**

b. A rain gauge station 'P' was inoperative for some time during a storm totals at 3-stations A,B, and 'C' Surrounding 'P' were 66.,48,0, and 37-mm respectively the normal annual precipitation at "P', A,B, and 'C' are 656.0,726.0518.0 and 382.0mm respectably. Estimate the storm precipitation at station 'P'. **06**

3. a. Explain Meyer's and Rohwer's methods for determination of evaporation. **07**

b. The data of average monthly percentage sunshine hours (p), mean monthly temperature (Tm) and vegetables crop coefficient (k) for a place at 20^0 N latitude is given **07**

Month	Aug	Sept	Oct	Nov
p	8.95	8.30	8.19	7.58
T_m°c(°F)	20(68)	15(59)	10(50)	11(51.8)
k	0.8	07	0.65	0.55

If a farmer grows vegetables from August to October, what is the total Evapotranspiration during this time period using Blaney-Criddle equation.

4. a. Define the term infiltration capacity of a soil. Explain Horton's infiltration capacity carve with a neat sketch. **08**

b. The rates of rainfall for a successive 30 minutes period of a 3-hr storm are 1.6, 3.6, 5.0, 2.8,2.2,1.0 cm/hr. The corresponding surface runoff is estimated to be 3.6 cm. Calculate ϕ-Index. **06**

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5. a. What are the various components of runoff? List the factors affecting runoff from a catchment. **07**

b. Compare the three empirical methods used for determining runoff. **07**

6. a. Describe with neat sketches, the seperation of base flow from run-off hydrograph. **06**

b. The ordinates of 6-hr unit hydrograph are given below. Derive a 12-hr unit hydrograph plot the unit 12-hr unit Hydrograph **08**

Time hr	0	3	6	9	12	15	18	21	24	27	30	33	36	39	42	45	48	51	54
wh $m^{3/s}$	0	200	500	1000	1600	2400	3500	4200	5200	4400	3100	2300	1500	1000	650	400	250	150	0

7. a. An engineer is required to design drainage system for a thermal power plant with an area of 2-0km^2 for a 50- year return period. The 50-year rain fall intensity in that region is given by, $I = \frac{38}{(t+11)^{0.38}}$, **07**

Where, I = Intensity in cm/hr

t = duration in minutes for time of concentration. If the time of concentration is 45 minutes for what discharge must the system be designed given the Runoff coefficient = 0.9.

b. Describe Muskingum's method of flood routing. **07**

8. a. Explain the following terms **08**

i. Confined Aquifer

ii. Cone of depression

iii. Porosity

iv. Aquiclude

b. A fully penetrating artesian well of 0.3 m diameter was pumped with a constant rate of 1800 m^3/day under a steady state condition. If the transmissivity of the aquifer is 0.145 m^2/min and the drawdown observed in the well is 8.92m, find the radius of influence of the well. **08**